

Vinh Ly

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Summary

Software Engineer with **18 years** of experience architecting and operating distributed systems at scale. Owned video streaming infrastructure processing **100+ million daily active users** and built payment systems handling **100,000+ transactions per second**. Designed and shipped backend services for Google Search serving billions of queries and YouTube playback pipelines processing millions of concurrent streams. Partnered with product, data science, and infrastructure teams to deliver end-to-end solutions. Seeking to bring deep systems expertise to a high-impact engineering team. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Languages: Go, Python, Java, C++, JavaScript, SQL, TypeScript, HTML, CSS

Infrastructure: Kubernetes, gRPC, Protocol Buffers, Redis, Memcache, Spanner

Databases: MySQL, PostgreSQL, Cassandra, Bigtable

Platforms: GCP, Linux, distributed systems, on-call

Work Experience

Staff Software Engineer | YouTube | 02/2025 - Present

- Designed video streaming infrastructure serving **100+ million daily active users** with sub-second latency requirements across global CDN edge nodes.
- Built playback pipeline components handling millions of concurrent streams using **Go** microservices communicating over **gRPC** and **Protocol Buffers**.
- Developed real-time monitoring and alerting systems for video quality metrics using **Python** and **Bigtable** for time-series data storage.
- Collaborated with client teams to define API contracts and ensure backward compatibility across video delivery platform versions.
- Optimized database query patterns in **Spanner** reducing read latency by 40% for high-traffic playback metadata endpoints.

Staff Software Engineer / Manager | Google | 03/2015 - 04/2024

- Led a team of 8 engineers building and maintaining core infrastructure for Google Search serving billions of daily queries across **Linux** clusters.
- Delivered migration of critical search serving paths from legacy **Java** monolith to **Go** microservices, improving deployment frequency from monthly to daily releases.
- Designed index serving infrastructure processing **thousands of queries per second** with p99 latency under 50ms using **C++** and custom memory allocators.
- Architected distributed caching layer using **Memcache** and **Bigtable** that reduced backend query load by 35% during peak traffic periods.
- Mentored 15+ engineers through performance review cycles and technical career development planning across junior to senior levels.

Software Engineer | Google | 02/2011 - 02/2015

- Built backend systems for ad auction infrastructure handling **millions of bid requests per second** using **C++** and **Protocol Buffers**.
- Reduced on-call incident resolution time by 25% through building automated failover systems and runbook-driven incident response tools.
- Developed data pipeline processing **terabytes of logs daily** using **MapReduce** and **Bigtable** for real-time auction analytics.

- Implemented feature flag and experimentation infrastructure serving **100+ million users** with sub-millisecond configuration propagation.

Software Design Engineer II | Microsoft | 10/2007 - 12/2011

- Developed enterprise backend services for **SQL Server** management tools using **C#** and **WCF** web services in **Windows** environments.
- Built data synchronization components between cloud and on-premise database instances handling **millions of records daily**.
- Designed and implemented automated testing frameworks reducing regression testing time by 50% for database migration workflows.
- Collaborated with product managers to define API specifications and ensure backward compatibility across service versions.

Software Design Engineer Intern | Microsoft | 06/2006 - 08/2006

- Developed prototype features for **Visual Studio** debugging tools using **C++** and the Windows debugging API.
- Conducted performance profiling and memory leak detection for large-scale C++ codebases under senior engineer guidance.

Education

B.S. Honors Degree in Computer Science

Virginia Tech | 2004 - 2007